

Exam_GRA4160_Spring2023_guidelines

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1 Exam Guidlines for GRA 4160 (Spring 2023)

1.1 Assignment 1 Guidelines:

1. Load the Data:
 - When loading data, make sure to import the required libraries and read the data from the appropriate source.
 - Since the dataset is distributed across four separate files, you need to concatenate these into a single DataFrame using the appropriate Pandas functions.
 - Be sure to inspect the structure and components of the DataFrame to ensure that the import and concatenation were done correctly.
 - a. Comment on the completeness of the dataset:
 - Use the appropriate Pandas functions to check for missing values in your DataFrame. Clearly report any missing values and their count.
 - Make comparisons across the four files in terms of data completeness.
 - Propose a suitable method for handling missing values based on the situation (for example, deleting missing data, replacing with the mean, median, mode etc.). Justify your chosen method.
 - b. Decide on the variable:
 - Consider whether all categories within the target variable are necessary for your analysis or if simplification (to binary classification, for example) is more suitable.
 - Justify your decision based on the nature of the dataset and the goals of your analysis.
2. Convert Categorical Features:
 - Identify the categorical variables in your dataset and use the appropriate Pandas methods to convert them to categorical data types.
 - Make sure to verify that the conversion was successful.
3. Feature Scaling:
 - Identify the continuous variables in your dataset that need to be scaled.
 - Choose a suitable feature scaling technique such as normalization or standardization. Justify your choice by discussing why it is suitable for your data and the problem at hand.
 - Apply the chosen scaling technique to the selected features and verify its successful application.
4. Feature Engineering:
 - a. Create New Features:
 - Use your understanding of the data and the problem at hand to construct new features that could potentially aid in predicting heart disease.

- Examples of such features could include age groups or other derivative variables based on existing data.
 - Plot the distribution of each new feature to verify its informativeness and relevance.
 - b. Transform Existing Features:
 - Consider transforming existing continuous variables into categorical ones if it aids in better representation or interpretation.
 - Make sure to justify the necessity of the transformation by explaining how it enhances the understanding of data or improves the model performance.
5. Exploratory Data Analysis (EDA):
- a. Data Visualization:
 - Use various visualization techniques (histograms, box plots, scatter plots, etc.) to plot the distribution of each feature.
 - Identify any outliers and examine relationships between features and the target variable. Make sure to provide your observations and interpretations of these plots.
 - b. Correlation Analysis:
 - Compute the correlation between features and visualize this using a correlation matrix and heatmap.
 - Identify any potential multicollinearity issues or redundant features. Be prepared to discuss how you would handle these issues.
6. Feature Selection:
- a. Feature Importance with Tree-Based Model:
 - Implement a tree-based model (such as a decision tree or random forest) to compute feature importance.
 - Identify and discuss the most relevant features based on the results.
 - b. Regularization-Based Methods:
 - Apply regularization techniques like Lasso or Ridge regression to the dataset. These methods are intended for continuous variables, if you work with a binary target variable, discuss how you would handle this.
 - Interpret the impact of these regularization techniques on feature selection. Identify and discuss the most relevant features based on the results of these techniques.

In each step of the process, you should aim to provide clear, concise code and commentary to illustrate your data handling decisions and techniques. This helps demonstrate your understanding and allows others to follow your process more easily. Remember that justifications for your choices are equally important as the choices themselves. Your reasoning should take into consideration the nature of the data, the objectives of the assignment, and best practices in data handling. Remember to provide clear and concise code, along with commentary on your observations and decisions throughout these processes. Your reasoning should reflect the nature of your data, the objective of the assignment, and best practices in data analysis.

1.2 Assignment 2 Guidelines:

1. Linear Regression:
 - Select suitable numerical features from the dataset as predictors to predict the age attribute using a linear regression model.
 - Justify the choice of features by explaining why they are suitable predictors for age.
 - Make sure to evaluate and interpret the model performance using appropriate metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute

Error (MAE), and R-squared.

2. Logistic Regression:

- Implement a logistic regression model to predict the target attribute.
- Select suitable features for this task, and explain why you selected these features. Your reasoning should take into account the nature of logistic regression and the requirements of the task.
- Evaluate the performance of your model using appropriate classification metrics like accuracy, precision, recall, F1 score, and ROC-AUC. Interpret the results.

3. Decision Tree Classifier:

- Use a decision tree classifier to predict the target attribute.
- Discuss how you configured the decision tree. This includes justifying your approach to feature selection, tree depth, splitting criterion, and other hyperparameters.
- Evaluate the performance of the decision tree model using appropriate classification metrics and interpret the results.

4. Model Evaluation:

- Evaluate the performance of all implemented models using appropriate metrics. For classification tasks (logistic regression and decision tree classifier), these could include accuracy, precision, recall, F1 score, and ROC-AUC. For the regression task (linear regression), consider using MSE, RMSE, MAE, and R-squared.
- Discuss why you selected these metrics and what they reveal about the strengths and weaknesses of your models. Your reasoning should reflect an understanding of these metrics and their suitability to the task at hand.

1.3 Assignment 3 Guidelines:

1. Simple Majority Vote Ensemble:

- Implement an ensemble consisting of a few decision trees. Each tree should be trained on a unique random subset of the training data.
- After training, use the ensemble to predict the labels of the test data.
- Discuss the concept behind the majority vote ensemble and why it could be effective for this task.

2. Bagging Ensemble:

- Construct a bagging ensemble of decision trees. Each tree should be trained on a distinct bootstrap sample from the training data.
- Use the ensemble to make predictions on the test data. -Provide insights into the concept of bagging and how it contributes to improving the performance of the model.

3. Random Forest Ensemble:

- Implement a random forest ensemble consisting of multiple decision trees, where each tree is trained on a different random subset of the features from the training data.
- Apply this ensemble to predict the labels of the test data.
- Discuss how the random forest ensemble differs from the bagging ensemble implemented earlier. Highlight the distinctive feature selection process in random forests and its potential benefits.

4. Boosting Ensemble:

- Construct a boosting ensemble and use it to predict the labels of the test data.
- Explain how this ensemble differs from the bagging and random forest ensembles implemented earlier. Discuss the core principle behind boosting and how it attempts to improve the model's performance.

5. Performance Comparison:

- Compare the performance of the four ensemble methods implemented above using appropriate metrics such as accuracy, precision, recall, F1 score, and ROC-AUC.
- Discuss the strengths and weaknesses of each method. Highlight any significant differences observed in their performance.
- Provide insights into why these models might perform differently, considering their different strategies for addressing bias-variance trade-off and improving model performance.

Remember to provide clear and concise code along with detailed commentary on your observations and decisions. Your reasoning should reflect an understanding of ensemble methods, their individual strengths and weaknesses, and how they apply to the task at hand.

1.4 Assignment 4 Guidelines:

1. Data Loading and Preprocessing:

- Load the Kaggle Telecom Churn dataset (Telco-Customer-Churn.csv) and understand its structure and variables.
- Clean and preprocess the data, which may include dealing with missing values, transforming categorical variables, and normalizing numerical features. Discuss the rationale behind each step you take.

2. Dataset Split:

- Split the dataset into a training set and a testing set. Justify your choice of the split ratio (e.g., why you chose a certain percentage for training vs testing).

3. Neural Network Design:

- Design a neural network model with one hidden layer. Describe the architecture you chose, including the activation function(s) used, the number of neurons in the hidden layer, and why you selected this particular structure.

4. Training Process:

- Train your neural network using the training set. Discuss the training process, including the loss function used, the optimization algorithm, and the performance metrics you chose to monitor.
- Explain why you chose these particular metrics and how they help in evaluating and guiding the training process.

5. Model Evaluation:

- Evaluate the performance of your neural network on the testing set using appropriate metrics (accuracy, precision, recall, F1 score, ROC-AUC, etc.).
- Interpret the results, highlighting the strengths and potential weaknesses of your model.

6. Reflection and Improvements:

- Reflect on the performance of your model. Discuss potential strategies to improve the performance of your neural network, such as tuning hyperparameters, changing the architecture, handling class imbalance, etc.
- Explain why you believe these strategies could enhance the performance.

7. Model Choice:

- Discuss whether a neural network is the best model for this problem. Justify your viewpoint, considering the strengths and limitations of neural networks, the nature of the data, and the task at hand.
- If you believe other models could be more suitable, discuss what these models might be and why you would consider using them.

Throughout your solution, ensure your code is clear, well-annotated, and replicable. Your commentary should reflect a solid understanding of neural networks and their application to the problem at hand, as well as best practices in data analysis and machine learning.